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1. Image Enhancement in the Spatial Domain

Image enhancement improves the quality of an image by modifying its pixel values directly.

$$\boxed{g(x,y) = T[f(x,y)]}$$

where,

$f(x,y)$ = Input image
$g(x,y)$ = Enhanced image
T = Transformation funtion

Techniques:

1. Grey - Level Transformations:

Modify the intensity values of pixels to improve brightness and contrast.

eg: Negative, log and Gamma transformations.

Users: Dark or low-brightness images.

2. Histogram Processing

Improve image contrast by redistributing gray level values using Histogram Equalization in an image

Types

↳ Histogram Equalization

↳ Histogram matching

uses: Low-contrast images where object details are difficult to identify

3. Spatial filtering

Applies a filter to neighbouring pixels.

types: → Smoothing filters

→ sharpening filter

Smoothing filters

↳ It is used to removing noise

By using, ⇒ Mean filter

⇒ Median filter

⇒ Gaussian filter

Sharpening filter:

↳ It is used to enhance edges and fine details.

By using, ⇒ Laplacian filter

⇒ Sobel filter

Advantages:

Improves image brighteness and contrast

Reduces image noise

Enhance edges and details.

Justification

Gray level transformation

→ It improves brightness and visibility

Histogram processing

→ It enhances contrast, making hidden details visible

Spatial filtering

→ It removes noise and sharpen edges, resulting in clearer images for object recognition and defect detection

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Image smoothing and image sharpening are two important image enhancement technique used in spatial domain smoothing removes unwanted noise, while sharpening enhances edges and fine details.

Difference between Image smoothing and Image sharpening

Image Smoothing	Image sharpening
Remove noise from the image.	Enhances edges and fine details
Uses low-pass filters	Uses high pass filters
Produces a smoother image.	Produce a sharper image
Eg: Mean, Median, gaussian filter	eg: laplacian, sobel, Prewitt filters.

Working Principle

1. Spatial smoothing filters

Smoothing filters replace each pixel with the average or median value of its neighboring pixels to red. noise

Mean filter formula: $g(x,y) = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} f(i,j)$

where,

$f(i,j) \longrightarrow$ input pixel values

$g(x,y) \longrightarrow$ smoothed image

$m \times n \longrightarrow$ filter size

2. Sharpening filters:

Sharpening filters enhance intensity changes make edges more visible.

Laplacian formula: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$

Application:

Smoothing filters	Sharpening filters
Medical imaging	Machine vision
CCTV images	Industrial inspection
Satellite images	Character recognition
Image preprocessing	Object detection.

∴ → Image smoothing improves image quality by reducing noise, while images sharpening improves clarity by enhancing edges and fine details. Both techniques are essential in image processing, and the choice depends on whether noise removal or edge enhancement is required.

3. Image Enhancement frequency domain

It improves image quality by processing the image after converting it from the spatial domain to the frequency domain using the Fourier transforma (FT). The image is filtered and the converted back using the Inverse fourier Transform (IFT).

Formula $$G(u,v) = H(u,v) \times F(u,v)$$

Where,

$F(u,v) \rightarrow$ Fourier transform of the input image

$H(u,v) \rightarrow$ Filter function

$G(u,v) \rightarrow$ Enhanced image in the frequency dom

Low-Pass filter (LPF).

Allows low-frequency components and blocks hig frequency components. Removes noise and smooth the ima

Applications: Medical imaging, satellite images, and image preprocessing.

High pass filter

Allows high frequency components and blocks low frequency components. Enhances edges and fine details

Application: Edge detection, industrial inspection, and defect detection

Comparison

Low-Pass filter (LPF)	High-Pass filter (HPF)
Removes noise	Enhances edges
Produces a smooth image	produces a sharp image
Reduces high-frequency components	Enhances high-frequency components.

$\therefore$ ⟹ Frequency domain enhancement improves image quality by filtering frequency components. Low-pass filters ares suitable for noise reduction and image smoothing, while high pass filters are suitable for edge enhancement and feature entraction.

4. Image Segmentation:

Image segmentation is the process of dividing an image into diffesent meaniful regions or objects.

Thresholding formula

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

where,

$f(x,y)$ → Input image

$g(x,y)$ → Segmented image

T → Threshold value

Types of segmentation

1. Point of Detection:

Detects isolated pixels that are significantly different from their surroundings.

2. Line Detection:

Detects straight lines in different diretions such as Horizontal, Vertical and diagonal.

3. Edge Detection:

Detects sudden change in intensity and identifies object boundaries. operators such as sobel, prewitt and canny are commonly used.

4. Thresholding

Separates an object from the background based on the pixel intensity

5. Region based segm...
Groups neighbouring pixels having similar p...
such as intensity or color

Suitable Technique:
For object and detect detection, edge detection and thresholding are most suitable technique.

5) Edge detection, Edge linking, boundary detection.
→ These are important Image processing techniques used to identify object shapes, recognize objects, and detect defects.

1. Edge Detection:
Detects sudden change in pixel intensity to identify object boundaries.
common operators: Sobel, Canny, Prewitt.

Gradient ~~formula~~: $|\nabla f| = \sqrt{G_x^2 + G_y^2}$

G_x → Horizontal gradient

G_y → Vertical gradient.

2. Edge linking:
Connects broken or disconnected edges to form continuous object boundaries.
Improves object recognition by creating complete shapes.

Boundary Detection:

Extracts the outer boundary of a segmented used to measure size & shape

$$\text{Boundary} = A - (A \ominus B)$$

where $A \rightarrow$ object

$B \rightarrow$ structuring element

$\ominus \rightarrow$ Erosion operation

$$\boxed{\text{Original image}} \downarrow \boxed{\text{Edge detection}} \downarrow \boxed{\text{Edge linking}} \downarrow \boxed{\text{Boundary detection}}$$